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System efficiency and power: the bridge between the device and system of a thermoelectric power generator†

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The thermoelectric efficiency formula ($\eta_{\max}$), derived by Ioffe, provides a direct connection between the material dimensionless figure of merit ZT and the maximum efficiency of transformation of heat into electricity in an ideal device with fixed terminal temperatures. However, for a thermoelectric power generator (TEG) system made for practical use, it could meet diverse temperature scopes and heat transfer conditions, making the terminal temperatures floating rather than fixed. Here, we define the optimized system efficiency and power output based on an analytical model that provides a connection between the device and the system by considering constant material properties and varying thermal circumstances. It is found that the dimensionless cold side thermal resistance f_c has a larger inhibiting effect on the system performance than its hot side counterpart f_h , suggesting more thermal management efforts should be placed on the cold side. Furthermore, a general system level design strategy is proposed to obtain the engineering leg length and the aspect ratio by considering the condition of external thermal resistance, output power and voltage match. As an example, a predicted 2–3 times enhancement was experimentally confirmed in a room temperature TEG device.

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Broader context

System level design of a thermoelectric power generator (TEG) is an interdisciplinary topic between thermal engineering and materials science. The conventional numerical approach to the thermal management of a TEG is effective but not instructive as a formula, such as the relationship between the device-level efficiency $\eta_{\max}$ and figure of merit ZT after Ioffe. We have developed an analytical framework for evaluating the system efficiency and output power of a TEG in practical service, taking into account the material properties, geometrical parameters and external heat transfer conditions. This theoretical work, together with experiment verification, provides the convenience to bridge the device and system of a TEG in a more direct way, and offers a white box solution for TEG system level design as compared with the conventional black box solution by numerical analysis, which would bring benefits for novel material and system level developments for the end-user's needs.

1. Introduction

Thermoelectric power generators (TEGs) can directly convert heat into electricity through the Seebeck effect of TE materials.

The TE device is definitely the core part of a TEG system, in which the output performance (the maximum conversion efficiency $\eta_{\max}$ and power density $w_{\max}$) is explicitly determined by the well-known formulae after Ioffe,¹ with given terminal temperatures and a constant property assumption as

$$\begin{cases} \eta_{\max} = \frac{\Delta T}{T_h} \frac{\sqrt{1 + Z\bar{T}} - 1}{\sqrt{1 + Z\bar{T}} + \frac{T_c}{T_h}}, & (a) \\ w_{\max} = \frac{\Delta T^2}{4L} PF, & (b) \end{cases} \quad (1)$$

where T_h , T_c , ΔT , and $\bar{T}$ are the hot side temperature, cold side temperature, temperature difference and average temperature, respectively. $Z\bar{T}$ is the dimensionless figure of merit at $\bar{T}$, L is

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the TE element length and PF is the power factor. Considering the temperature dependent transport properties of most TE materials, the above formulas could lead to unreliable predictions of the TE device performance under large temperature differences. Several studies^{2–9} have been performed to solve this problem by deriving analytical expressions or numerical methods for the maximum efficiency. Among them, the engineering ZT method proposed by Kim *et al.*⁸ and the reduced current density method by Snyder and Ursell² are established on the least simplifications and give comparably accurate predictions of the TE device efficiency over wide ranges of TE materials and temperature differences.¹⁰

Furthermore, in a real application, the validity of fixed terminal temperatures (*i.e.*, T_h and T_c) needs to be reconsidered. For a TEG in operation, a heat flow from the hot side to the cold side must be established. To realize this, some kind of heat transfer medium is essential to maintain thermal contact between the TE module and heat reservoirs. Typical heat transfer media include air or water heat exchangers for waste heat recovery scenarios in automobiles,^{11,12} absorbers for solar thermoelectric generators (STEGs)^{13,14} or some industrial processes with high temperatures,^{15,16} and radiators for radioisotope thermoelectric generators (RTGs) for space applications.^{17,18} Apart from the TE elements and heat exchangers, the substrates and electrodes inside the TE module as well as the micro gaps or thermal interfacial materials^{19,20} between the TE module and heat exchangers also introduce certain levels of thermal resistance into the system. Analogous to an electric circuit, the effective temperature drop across the TE elements is determined by the thermal resistance distribution in the whole system and cannot be known *a priori*. Moreover, the equivalent thermal resistance of the TE elements varies with the electric current,^{21–24} which makes the situation more complex. Consequently, eqn (1) is insufficient to give a sound evaluation of a TEG's output performance in realistic thermal circumstances.

Owing to the coupling interaction between the thermal and electrical networks and the induced floating nature of the effective temperature range within the thermoelectric legs, it is essential to simultaneously perform thermal and electrical analyses on the whole system. Numerical methods, *i.e.*, finite element analysis, are proven to be an effective way to solve this kind of complex problem, with all the influence factors, including the temperature dependent properties, the device geometries and external heat exchanger parameters, being taken into account in an integrated model.^{25–30} However, it is rather computationally costly, especially for TEG systems with massive legs. Moreover, the solution based on numerical simulations is more like a black-box, which can't provide concise and insightful guidelines as analytical formulae do. On the other hand, efforts have also been made to analytically solve this problem, while no general analytical formula of the system efficiency or power output has been given yet. Existing analytical work is mainly based on simplifications such as neglecting the effect of electric current on the effective temperature range^{31,32} or ignoring the variations of the heat flow within the whole system,^{33–36} making the models only valid under limited temperature differences and external thermal resistances.

Gao and Rowe³⁷ developed a model for TEGs to account for the thermal and electrical contact resistances caused by the substrates and electrodes, with intuitive expressions for the reduced effective temperature difference and power output as

$$\begin{cases} \Delta T = \frac{\Delta T_o}{1 + 2rw}, & (a) \\ P = \frac{P_i}{(1 + 2\rho_c/(\rho L_o))(1 + 2rw)^2}, & (b) \end{cases} \quad (2)$$

where T_o is the total temperature difference across the module, and P_i is the ideal power output neglecting the electrical and thermal contact resistances. r and w are respectively the thermal conductivity ratio and thickness ratio between the TE material and the substrate, and hence their product rw represents the relative increase in the thermal resistance caused by the existence of the substrates. ρ_c and ρ are respectively the electrical contact resistivity and the TE material electrical resistivity, and L_o is the TE leg length, and thus $\rho_c/(\rho L_o)$ represents the contribution of contact effects to the electrical resistance. Since this model only considers the contact resistances inside the TE module, it can be used to evaluate the device performance only if the hot and cold side temperatures of the module are measured. Nevertheless, the influence of thermal resistances between the module and the heat source (sink) could not be taken into account. In Stevens' work on TEG thermal design under small temperature difference (1–10 K) conditions,³¹ it was found that the temperature difference equipartition (also noted as the thermal impedance match) condition was directly responsible for maximum power generation. Mayer and Ram²¹ discussed the optimization of heat sink-limited TEGs. The optimum electrical load was for the first time claimed to be larger than that in ideal cases (equal to the internal resistance) for maximum power output in realistic thermal environments. An approximate explicit expression ($L_{opt} \approx \kappa A_{TE}/(hA)$) was given according to the thermal resistance match condition, based on a pure thermal conduction assumption within the TE elements. The superior role of the thermal conductivity (κ) over the Seebeck coefficient and the electric conductivity for higher efficiency and power output was highlighted in heat sink limited situations.

In order to analyze the heat flow and temperature distribution within a TEG in a convenient but electrical current dependent manner, several efforts have been made to integrate the thermal conductive term and the convective terms into a single equivalent thermal conductivity or conductance. Baranowski *et al.*²³ recommended the same expression for optimum leg length for maximum power output as given in Mayer and Ram's work,²¹ while the material thermal conductivity was replaced by an effective thermal conductivity, which was first proposed in their work based on thermoelectric compatibility theory.² Another concept of effective thermal conductance for a single TE leg explicitly including the load information was adopted by Apertet *et al.*^{35,36} and McCarty,³⁴

$$K_{eff} = \frac{\kappa A}{L} \left(1 + \frac{ZT}{1+m} \right) \quad (3)$$

where m denotes the ratio of the load resistance to the internal electrical resistance of the TE module. It should be emphasized that

the effective thermal conductance needs to be used under the constant heat flux assumption within the whole system, which is valid only under limited temperature differences. There are some other semi-analytical models^{22,38–40} that abandon the above simplifications, and thus remove the limitations on the application range; however, they are always relying on iterative methods to give quantitative results and no closed form of analytical solution is obtained.

In this work, we define the optimized system efficiency and power output by a set of explicit formulae in realistic thermal environments, which are derived from an integrated system level model. A direct connection between the system performance and the device parameters as well as the external thermal conditions is for the first time established, showing a clear physical picture for a TEG system operating with varying thermal circumstances. We have obtained the optimum electrical resistance match conditions for maximum system efficiency and output power, and suggest that they highly depend on the thermal resistance distribution. A system level enhancement in the output power of a room temperature TEG is theoretically predicted and experimentally validated under practical thermal boundaries. Moreover, we provide a general strategy for the system level design of TEGs to achieve the target output power and voltage densities by tuning a newly defined dimension parameter, the engineering leg length, and the aspect ratio of the TE legs.

2. System efficiency and output power

In order to evaluate the system level performance, we construct an integrated TEG system, comprised of the TE device, heat reservoirs and the heat transfer media between them, as shown in Fig. 1(a). Fig. 1(b) presents the thermal resistance network corresponding to the TEG system with a heat source temperature of T_h and a heat sink temperature of T_c . For simplicity, only one TE leg is considered in the analysis and the fill factor is set to one. The influence of a practical fill factor can be easily reflected in the dimensionless thermal resistance parameters f_h and f_c , which are defined as ratios of the hot side ($R_{t,h}$) and the cold side ($R_{t,c}$) thermal resistances to the conduction-induced thermal resistance ($R_{t,l}$) of the TE leg, respectively. As shown in Fig. 1(b), the effective temperature difference ($\Delta T'$) contracts from the total available temperature difference (ΔT) due to the

existence of external thermal resistances as well as the electrical current flowing through the TE leg in service. The TE leg terminal temperatures at the hot and cold side are denoted as T_h' and T_c' , correspondingly.

Firstly, we will derive the explicit expression for the effective temperature difference $\Delta T'$ across the TE leg, which is done based upon the fundamental heat transfer analysis of the system. Under one-dimensional electric current and energy flow circumstances, the temperature distribution and heat flux within the TE leg in the steady state are characterized by (see the ESI,[†] Derivation of the governing equation of the thermo-electric process, for a detailed derivation)

$$\begin{cases} \nabla \cdot (\kappa \nabla T) + \frac{J_e^2}{\sigma} - \tau \vec{J}_e \cdot \nabla T = 0, & (a) \\ \vec{q} = \alpha T \vec{J}_e - \kappa \nabla T, & (b) \end{cases} \quad (4)$$

where α , σ , κ , τ , and J_e are the Seebeck coefficient, electrical conductivity, thermal conductivity, Thomson coefficient, and the electric current density, respectively. When further assuming temperature independent physical properties (α , σ , and κ), the Thomson coefficient becomes zero, and the governing equation and boundary conditions of the TE leg in the steady state are given by

$$\begin{cases} \kappa \frac{d^2 T}{dx^2} + \frac{I^2}{\sigma A^2} = 0, & (a) \\ -\kappa A \frac{dT}{dx} + \alpha T_h' I = \frac{T_h - T_h'}{R_{t,h}}, x = 0, & (b) \\ -\kappa A \frac{dT}{dx} + \alpha T_c' I = \frac{T_c' - T_c}{R_{t,c}}, x = L, & (c) \end{cases} \quad (5)$$

where I , A and L are the electric current, cross-sectional area and length of the TE leg, respectively. An analytical solution of the temperature T with respect to the electric current I is obtained (see the ESI,[†] eqn (S21)), leading to an exact expression of the effective temperature difference as

$$\Delta T' = \frac{\Delta T - \alpha R_{t,l} I (f_h T_c + f_c T_h + f_h f_c I^2 R R_{t,l}) + 0.5 I^2 R R_{t,l} (f_h - f_c)}{f_h + f_c + (1 + f_h \alpha R_{t,l} I) (1 - f_c \alpha R_{t,l} I)} \quad (6)$$

where R is the internal electrical resistance of the TE leg. Moreover, in power generating mode the electric current is governed by the effective temperature difference in the following manner,

$$I = \frac{\alpha \Delta T'}{R(1+m)} \quad (7)$$

where m is the ratio of the load resistance to the TE leg electrical resistance. Theoretically, the electric current I as well as the effective temperature difference $\Delta T'$ could be solved for with respect to the electric resistance ratio m by substituting eqn (6) into eqn (7), which results in a cubic equation of the electric current I (see the ESI,[†] eqn (S24)). Unfortunately, the cubic equation of the electric current is too complex to be analytically solved, which makes it impossible to carry out

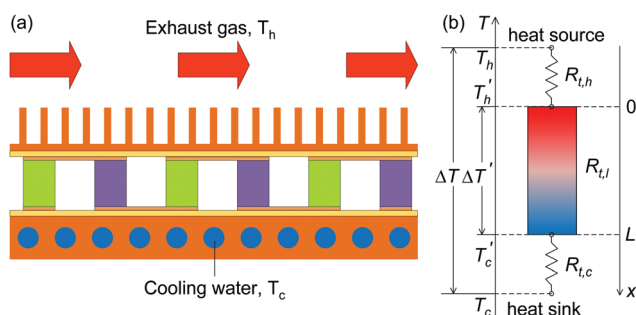


Fig. 1 A typical integral TEG system. (a) Waste heat recovery TEG system; and (b) thermal network of a TEG system.

further derivations of the system efficiency and output power. To address this problem, a simplified expression for the effective temperature difference is given by

$$\Delta T' = \frac{\Delta T - \alpha R_{t,l} I (f_h T_c + f_c T_h)}{1 + f_h + f_c} \quad (8)$$

in the first step to continue the derivation, and the validity of the simplification in eqn (8) will be checked later on. Based on eqn (8), the explicit formulas of maximum conversion efficiency and power output per leg area are derived as follows, (see the ESI,[†] Derivation of the maximum efficiency and power formula, for a detailed derivation)

$$\begin{cases} \eta_{\max} = \frac{\Delta T}{T_h} \frac{m_{\text{opt},\eta} - 1}{m_{\text{opt},\eta} + \frac{T_c}{T_h} + \beta}, & \text{(a)} \\ w_{\max} = \frac{1}{(1 + f_h + f_c)^2} \frac{\Delta T^2}{4 L m_{\text{opt},w}} \text{PF}, & \text{(b)} \end{cases} \quad (9)$$

where $\beta = 2 \frac{T^*}{T_h} \frac{\bar{T}}{T_h} m_{\text{opt},\eta} \left(1 + \frac{m_{\text{opt},\eta}}{1 + Z T^*} \right)$, $T^* = \frac{f_h T_c + f_c T_h}{1 + f_h + f_c}$,

$\bar{T}^* = \frac{\bar{T}}{1 + f_h + f_c}$, and $\text{PF} = \alpha^2 \sigma$.

Eqn (9) is similar in form to the ideal formulas listed in eqn (1), so it can be regarded as a concise description of the system level performance of a TEG operating in realistic thermal environments. The only difference between the efficiency formulas in eqn (9(a)) and eqn (1(a)) lies in the addition of a dimensionless term β in the denominator of the second multiplier for a comprehensive account of the effects of realistic thermal conditions, i.e., the non-zero f_h and f_c . The maximum system power output in eqn (9(b)) is evaluated by multiplying by a correction factor $\left(\frac{1}{(1 + f_h + f_c)^2} \right)$ in the original form in eqn (1(b)). Meanwhile, the optimal conditions for maximum efficiency and power output are found to deviate from the ideal cases as

$$\begin{cases} m_{\text{opt},\eta} = \sqrt{(1 + Z T^*) [1 + Z (T^* + \bar{T}^*)]}, & \text{(a)} \\ m_{\text{opt},w} = 1 + Z T^*, & \text{(b)} \end{cases} \quad (10)$$

Both the terms T^* and $\bar{T}^*$ have a unit of temperature (K), which could adjust the optimum electrical conditions according to the external heat transfer conditions. In an ideal heat exchange condition, i.e. $f_h = f_c = 0$, T^* becomes 0 while $\bar{T}^*$ returns to $\bar{T}$, respectively. Thus $m_{\text{opt},\eta} = \sqrt{1 + Z \bar{T}}$, $m_{\text{opt},w} = 1$, and eqn (9) becomes identical to eqn (1). This consistency implies the essential rationality in the proposed explicit formula. In contrast, as the external thermal resistances f_h and f_c increase at the same speed, T^* increases and levels off at $\bar{T}$, while $\bar{T}^*$ gradually diminishes to 0. As a result, both $m_{\text{opt},\eta}$ and $m_{\text{opt},w}$ approach the same value $1 + Z \bar{T}$, which is a special case in agreement with the deduction of Apertet *et al.*^{35,36}

The proposed explicit formulas are validated using a numerical method over wide ranges of total temperature differences (50–500 K) and external thermal resistance levels (thermal resistance ratios between 0.01 and 10) covering most practical

application cases (see the ESI,[†] Validation of the TEG model, for detailed information). The analytical results of maximum efficiency and power output as well as the corresponding electrical resistance ratios agree well with those from numerical computations (deviations within $\pm 1.75\%$) under a moderate total temperature difference of 50 K and a wide thermal resistance ratio range. Under a temperature difference higher than 500 K, the efficiency formula is still capable of giving satisfactory predictions (deviations within $\pm 5\%$) for all of the considered external heat transfer conditions, while it is noted that a larger deviation up to 16% occurs for the power output formula compared to the numerical data under asymmetric thermal resistance distributions.

The relatively higher errors of the output power stem from the simplifications made in the effective temperature difference formula eqn (8). To make it more clear, the error distribution of the effective temperature difference evaluated by eqn (8) is investigated with respect to the thermal resistance parameters. Furthermore, a comparison is made between eqn (8) and two existing expressions for the effective temperature difference in the literature^{33,35} (see the ESI,[†] Modification of the TEG model, for detailed information),

$$\Delta T' = \frac{\Delta T - \alpha R_{t,l} I (f_h + f_c) T_c}{1 + f_h + f_c} \quad (11)$$

$$\Delta T' = \frac{\Delta T - \alpha R_{t,l} I (f_h + f_c) \bar{T}}{1 + f_h + f_c} \quad (12)$$

In the above two equations, the one proposed by Apertet *et al.*³⁵ (eqn (12)) is found to be in a good complementary relation with eqn (8). Firstly, both eqn (8) and (12) work well when the external thermal resistance parameters (f_h and f_c) are lower than 0.2 (see the ESI,[†] Fig. S3 and S4). Secondly, in the range of 0.2–2 for f_h and f_c , both eqn (8) and (12) deviate from the exact values of the effective temperature difference, but in the opposite direction, which is important for the following weighted summation method. Last, when f_h and f_c are higher than 2, eqn (8) recovers its good capability, while eqn (12) continues to degrade. Considering the behaviors of eqn (8) and (12), a weighted summation is carried out between these two formulas, with a weight ratio of $f_h + f_c$ for eqn (8) and 1 for eqn (12), based on which a new formula of the effective temperature difference $\Delta T'$ is given by

$$\Delta T' = \frac{\Delta T}{1 + f_h + f_c} - \frac{\alpha R_{t,l} I (f_h + f_c) [\bar{T} + (f_h T_c + f_c T_h)]}{(1 + f_h + f_c)^2} \quad (13)$$

On the basis of eqn (13), a new formula for the maximum output power is derived as

$$\begin{cases} w_{\max} = \frac{1}{(1 + f_h + f_c)^2} \frac{\Delta T^2}{4 L m_{\text{opt},w}} \text{PF} & \text{(a)} \\ m_{\text{opt},w} = 1 + \frac{f_h + f_c}{1 + f_h + f_c} Z (T^* + \bar{T}^*) & \text{(b)} \end{cases} \quad (14)$$

It is noticed that the power output formula is the same as that in eqn (9), and the only change occurs in the optimum electrical resistance ratio. This new formula is validated to give

reliable estimations (deviations within $\pm 4\%$) of the maximum power output throughout widely experienced thermal circumstances (see the ESI,† Fig. S5), and thus a general set of formulas for the reliable estimation of the optimal conversion efficiency and power output of a real TEG system is finally obtained as

$$\begin{cases} \eta_{\max} = \frac{\Delta T}{T_h} \frac{m_{\text{opt},\eta} - 1}{m_{\text{opt},\eta} + \frac{T_c}{T_h} + \beta} & \text{(a)} \\ m_{\text{opt},\eta} = \sqrt{(1 + ZT^*)[1 + Z(T^* + \bar{T}^*)]} & \text{(b)} \\ w_{\max} = \frac{1}{(1 + f_h + f_c)^2} \frac{\Delta T^2}{4Lm_{\text{opt},w}} \alpha^2 \sigma & \text{(c)} \\ m_{\text{opt},w} = 1 + \frac{f_h + f_c}{1 + f_h + f_c} Z(T^* + \bar{T}^*) & \text{(d)} \end{cases} \quad (15)$$

In the current work, we aim at offering a concise and general formula which gives a clear physical picture of the TEG system performance. Owing to this, several simplifications are made, which should be kept in mind when applying the formulae in eqn (15). Firstly, the TE material properties are assumed to be temperature independent, which could limit the application of the formula in cases with large available temperature differences and significantly temperature dependent materials. Secondly, both the heat source and the heat sink temperature are assumed to be constant, which is suitable for cases with moderate heat fluxes such that the heat source/sink temperatures would not be severely influenced. More analytical efforts are needed to handle TEG applications with varying heat source/sink temperatures, which will be our subject in the next step. Thirdly, the thermal

radiation between the substrates at the hot and cold sides is recognized as a secondary effect and neglected, since it could play an important role only under a large temperature difference. Furthermore, it could be effectively suppressed by using substrates or coatings with low emissivities. In spite of the above simplifications, we will show that eqn (15) is already a useful analytical tool for many common TEG applications, including waste heat scenarios with temperature differences within 200 K and human body energy harvesting applications.

Here, we apply the proposed formula to calculate the maximum power output of a chimney TEG numerically investigated in ref. 25 and 26 under various external heat exchanger designs. Fig. 2(a) and (b) present comparisons of the power output per leg predicted by the analytical method in this study and numerical data from ref. 25 and 26. Also shown in Fig. 2(c) and (d) are the corresponding thermal resistance ratios for each heat exchanger design as well as the heat source and sink temperatures. In the numerical simulations in ref. 25 and 26, the temperature dependent material properties were taken into account, while in the analytical evaluations given by the currently proposed model, the average value of each property within the total temperature range is used. Reasonable agreements are achieved between the analytical predictions and numerical data, confirming that the formula can be used to conveniently estimate the system performance of practical TEG applications under temperature differences within 200 K. Some local deviations between the analytical and numerical results, *e.g.*, for the 7th heat exchanger design in Fig. 2(a) and the 3rd and 23rd heat exchanger designs in Fig. 2(b), are mainly caused by the average treatment of the heat reservoir temperatures or by the neglect of the temperature dependence of the material properties.

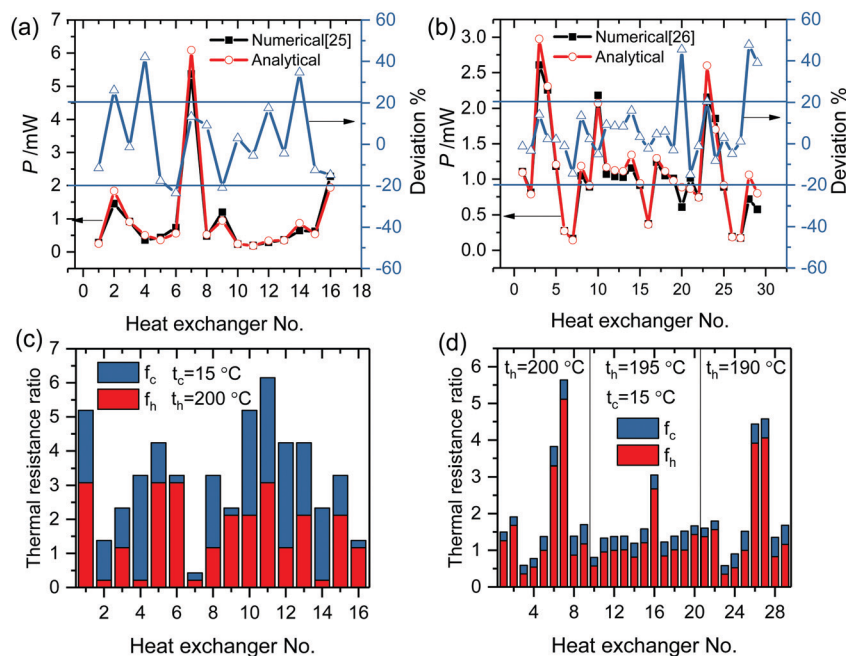


Fig. 2 (a) and (b) Comparisons of analytical and numerical results for maximum power of a TEG for waste heat recovery under different heat exchanger designs; and (c) and (d) specific thermal resistance ratios for each heat exchanger design and heat reservoir temperatures. Each heat exchanger No. represents a heat exchanger with specific design parameters, *e.g.*, fin height, fin gap, etc.

3. Effect of the external thermal resistance

3.1 Specific thermal resistance parameters, f_h and f_c

To conduct an estimation of the range of external thermal resistance parameters f_h and f_c in realistic thermal circumstances, we consider a TE leg with typical values of physical properties and geometric dimensions as listed in Table 1. Table 2 illustrates the typical heat transfer coefficients of some of the most commonly used heat transfer media for TEGs. The thermal resistances of the TE leg and the typical external heat transfer media are evaluated by $R_{t,l} = L/\kappa A$ and $R_{t,ex} = FF/(hA)$, respectively, in which FF is the fill factor. It is seen that the typical thermal resistance ratios are within 0.045–9. When further considering the thermal resistances of the substrates, electrodes and solder layers within the TE module, the total external thermal resistance ratio at each side of a TE element is expected to cover a range of 0.05–10 in most cases.

3.2 Effect of the thermal resistance distribution

Firstly, the effect of asymmetric distributions of thermal resistances at the hot and cold sides is investigated. Fig. 3(a) and (b) respectively present the evolutions of the maximum system efficiency and output power under two critical thermal resistance distributions, *i.e.*, one with zero thermal resistance at the hot side and the other with zero thermal resistance at the cold side. Meanwhile, the relative deviation curves between the two asymmetric thermal resistance distributions under different total temperature differences are shown in Fig. 3(c) and (d). Under a relatively low total temperature difference of 50 K (only illustrated in Fig. 3(c) and (d)), the deviations of the maximum system efficiency and output power under the above two thermal resistance distributions are well limited within 10% throughout the whole range of the thermal resistance parameter. Overall, the thermal resistance distribution does not have a significant impact on the system performance under low total temperature differences. Under higher total temperature differences, *i.e.*, 250 K and 500 K, the deviations are still trivial when the thermal resistance parameter at the non-zero side is lower than 0.1, while for higher thermal resistance parameters, the deviations between two asymmetric thermal resistance distributions are significantly magnified. As an example, under

a total temperature difference of 500 K and a thermal resistance parameter of 1, the maximum system efficiency and output power with zero thermal resistance at the cold side are 35% and 21% higher than those with the opposite thermal resistance distribution, respectively, and the deviations further rise over 60% with the thermal resistance parameter increasing to 10. Meanwhile, the optimum electrical resistance matching parameters for the two critical cases also experience distinct variations under relatively large total temperature differences and thermal resistance parameters (see the ESI,† Effect of the thermal resistance distribution, Fig. S6, for detailed information). Therefore, we demonstrate that the external thermal resistance at the cold side of a TEG system has a greater impact on the system performance compared to that at the hot side. This phenomenon can be explained directly by the proposed analytical model. According to the effective temperature difference expressions (eqn (8) and (13)), the cold side thermal resistance parameter f_c is multiplied by the heat source temperature T_h , while the hot side thermal resistance parameter f_h is multiplied by the heat sink temperature T_c . Here T_h and T_c can be regarded as weight coefficients of f_c and f_h . Since T_h is always higher than T_c , f_c plays a more important role than f_h , and this trend is more pronounced with a larger difference between T_h and T_c . Consequently, when implementing heat transfer improvements of a TEG system for enhanced output performance, the heat transfer condition at the cold side should be considered with a higher priority, particularly for cases with relatively large available temperature differences and external thermal resistances.

3.3 Effect of the thermal resistance level

Some researchers^{35,36} have already pointed out that the electrical operating conditions for maximum conversion efficiency and power output could coincide in specific situations with high external thermal resistance levels. Here we revisit this situation by using a combination of a numerical scanning method and the analytical model presented in this work, and try to uncover more useful information for the optimal operations of TEGs. Fig. 4(a) and (b) display the numerically obtained evolution curves of the conversion efficiency and power output of a TE leg *versus* the electrical resistance ratio m under different symmetric thermal resistance levels and a constant total temperature difference of 50 K. Both $m_{opt,\eta}$ and $m_{opt,w}$ increase with the external thermal resistance, as illustrated in Fig. 4(c).

Also shown in Fig. 4(c) are the upper and lower limiting lines for 99% η_{max} and 99% w_{max} , respectively presented by the dash dot lines and dash lines. Apertet *et al.* in their earlier work³⁵ suggested taking the region lying between the two optimal m curves (the blue region marked in Fig. 4(c)) as the optimal selection range to achieve a balance between efficiency and power. They demonstrated a reduction in the optimal selection

Table 1 Parameters of a typical TE leg

Physical properties	Values	Geometric dimensions	Values
α ($\mu\text{V K}^{-1}$)	200	Length (mm)	2
σ (S cm^{-1})	1000	Width (mm)	2
κ ($\text{W m}^{-1} \text{K}^{-1}$)	1.5	Depth (mm)	2

Table 2 Thermal resistance estimations of the TE leg and typical heat transfer media

	TE leg	Air convection (no fin)	Air convection (enhanced fins)	Water convection	Water phase change
Heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)	—	25	250	2000	5000
Thermal resistance (K W^{-1})	333.3	3000	300	37.5	15
Thermal resistance ratio	—	9	0.9	0.11	0.045

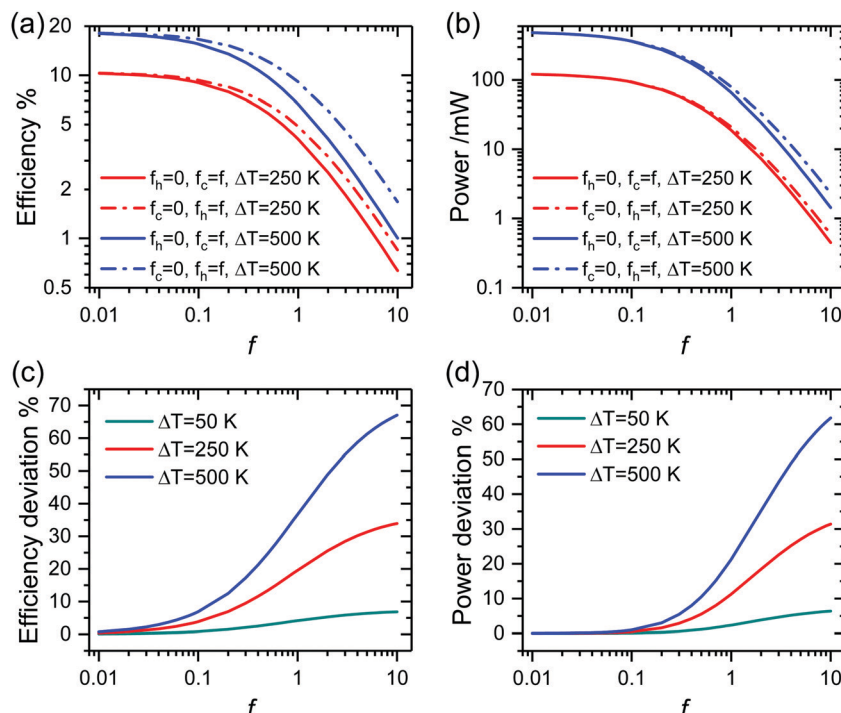


Fig. 3 (a) Evolution curves of maximum system efficiency for two critical thermal resistance distributions, one with zero thermal resistance at the hot side, and the other with zero thermal resistance at the cold side, with f denoting the non-zero side thermal resistance parameter; (b) evolution curves of the maximum output power for two critical thermal resistance distributions; (c) deviations of the maximum system efficiency between two critical thermal resistance distributions; and (d) deviations of the maximum output power between two critical thermal resistance distributions under different temperature differences.

range with increasing external thermal resistance. Meanwhile, from another perspective, the m range for simultaneously realizing power and efficiency higher than 99% of their respective maximum values as marked by the red region is remarkably augmented and significantly overwhelms the blue region under higher external thermal resistances. Since 99% of the optimum performance is a rather satisfying level, the system is indeed offered with more adjustment room for the electric load in harsh thermal environments, although the output performance is inevitably reduced. Considering this, the summation of the blue and red regions is believed to represent a better optimal selection range of m compared to the blue region alone, accounting for different external thermal resistance levels.

Fig. 4(d) presents the relative enhancement of the newly optimized system power by using eqn (15c and d) in comparison to the conventional electrical resistance matching method (*i.e.*, $m_{\text{opt,w}} = 1$) with varying total temperature differences and external thermal resistance levels (of a symmetric distribution). It gives us clear knowledge of how much benefit in the output power of a TEG we can get by using the newly proposed optimization formula in realistic thermal environments. For external thermal resistance ratios below 0.1, an ignorable enhancement could be noticed, while for relatively higher external thermal resistance levels, the difference becomes considerable. As an example, under a total temperature difference of 500 K, the enhancement in output power by using the newly proposed optimization formula reaches 10% for a thermal resistance

parameter of 1 at both sides, and finally approaches 20% for higher thermal resistance parameters. Hence, without any change in the heat exchangers and the device parameters, visible rises in the output power could be realized just by updating the electrical match condition according to eqn (15d).

4. TEG system design

4.1 Power density

Here we consider the design procedure of TEGs with limited available temperature difference, which is the common case for ambient energy harvesting techniques. As an example, for TEGs used for human body energy harvesting, the available temperature difference is in most cases well limited within 20 K.⁴⁵ Moreover, since the available area is also limited, the power density is of critical significance. To clarify the relationship between the design parameters and the power density, a general formula is derived based on the proposed model as (see the ESI,[†] Design of TEGs with limited T , for a detailed derivation)

$$P_{\text{dens}} = \frac{1}{4} Z \Delta T^2 \frac{1}{\frac{L^*}{\kappa} + \frac{\kappa}{L^*} \left(\frac{1}{h}\right)^2 \left[1 + Zh \left(\frac{T_c}{h_h} + \frac{T_h}{h_c}\right)\right] + \frac{1}{h} (2 + Z\bar{T})} \quad (16)$$

where L^* is defined as the ratio between the TE leg length and fill factor, *i.e.*, $L^* = \frac{L}{\text{FF}}$, while the total heat transfer coefficient

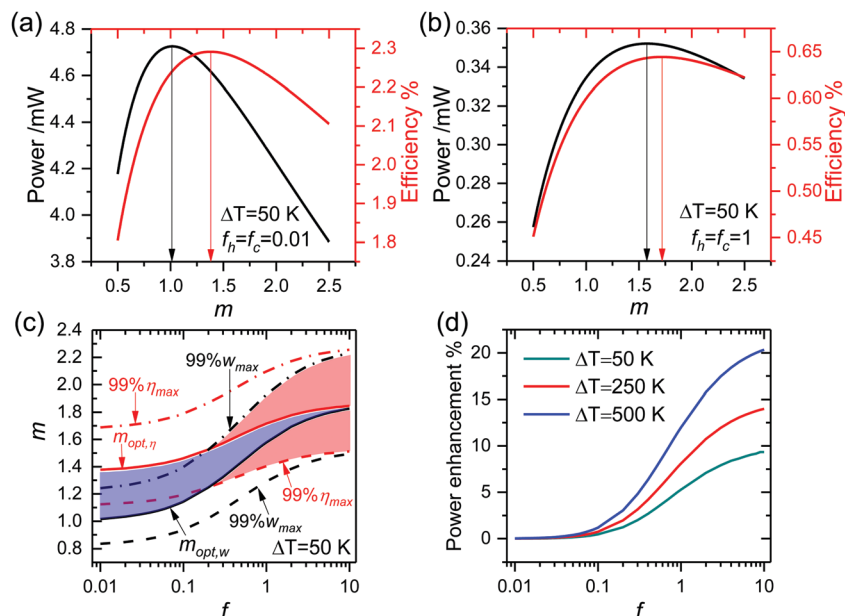


Fig. 4 (a and b) Evolutions of the system performance with varying m , $\Delta T = 50$ K; (c) optimum m for maximum efficiency and power with varying external thermal resistance level; and (d) enhancement of the newly optimized system powers over the conventionally obtained ones in realistic thermal environments.

h holds the relation of $\frac{1}{h} = \frac{1}{h_h} + \frac{1}{h_c}$ with the heat transfer coefficient at the hot side (h_h) and cold side (h_c). Eqn (16) suggests that the power density is determined by three aspects, *i.e.*, the material properties (Z and κ), the dimension parameters (lumped as L^* , named the engineering leg length here) and the external heat transfer conditions ($\frac{1}{h}$, $\frac{1}{h_h}$ and $\frac{1}{h_c}$). Considering a given thermoelectric material while changeable leg dimension parameters, the optimum relationship between the engineering leg length and the heat transfer conditions for maximum power density is given by

$$L_{opt}^* = \frac{\kappa}{h} \sqrt{1 + Zh \left(\frac{T_c}{h_h} + \frac{T_h}{h_c} \right)} \quad (17)$$

When considering the small temperature difference condition, an approximation could be made as $T_h \cong T_c \cong \bar{T}$, and hence eqn (17) turns into a simpler form

$$L_{opt}^* = \frac{\kappa}{h} \sqrt{1 + Z\bar{T}} \quad (18)$$

which is equivalent to the optimal TE leg length formula by Apertet *et al.*³⁶ and similar to the optimization rule for maximum power density by Baranowski *et al.*,²³

$$\frac{L}{FF} = \frac{\kappa_{eff}}{h} \quad (19)$$

where κ_{eff} is named the effective thermal conductivity, which is calculated based on thermoelectric compatibility theory as

$$\kappa_{eff} = \frac{\kappa T_h' (1 + zT + \sqrt{1 + zT})}{2(T_h' - T_c')} \left[1 - \left(\frac{T_h'}{T_c'} \right)^{k_g} \right] \quad (20)$$

where $k_g = \frac{2 - 2\sqrt{1 + zT}}{zT}$. In the reference, it was claimed that the value of the effective thermal conductivity does not depend on the operating conditions, but only depends on the intrinsic properties of the thermoelectric materials. In fact, for TEGs with limited available temperature differences, the deviation between T_h' and T_c' is expected to be rather small. In this situation, κ_{eff} can be proven to be nearly equal to $\kappa\sqrt{1 + Z\bar{T}}$ with a reasonable approximation as (see the ESI,[†] Design of TEGs with limited T , for detailed information)

$$1 - \left(\frac{T_h'}{T_c'} \right)^{k_g} \cong k_g \left(1 - \frac{T_h'}{T_c'} \right) \quad (21)$$

From the above discussion it can be stated that eqn (16) and (17) can be used as a general guideline for achieving maximum power density, which turns into a simpler form of eqn (18) under small temperature difference conditions, and is found to encompass the design rules proposed by Apertet *et al.*³⁶ and Baranowski *et al.*²³

Although eqn (18) gives the optimum engineering leg length L_{opt}^* for maximum power density, it should be pointed out that the real engineering leg length is often constrained by some aspects accounting for structural stability, flexibility and other requirements, making it difficult to reach the optimum value. Generally, a larger L^* means a longer leg length or a smaller fill factor, which is susceptible to more stability and reliability issues. To discuss this issue, we start with a commercially typical TE device for human body energy harvesting,⁴⁵ whose parameters and thermal boundary conditions (denoted as the initial case) are listed in Table 3. As shown in Fig. 5(a), the engineering leg length L^* is 4 mm in the initial case ($L = 1$ mm,

Table 3 Initial TEG parameters and operating conditions

Parameter	Value	Parameter	Value	Parameter	Value
Seebeck coefficient α ($\mu\text{V K}^{-1}$)	200	Leg length L (mm)	1	Air side heat transfer h_c ($\text{W m}^{-2} \text{K}^{-1}$)	100
Electrical conductivity σ (S cm^{-1})	1000	Fill factor FF	0.25	Skin temperature T_h ($^{\circ}\text{C}$)	34
Thermal conductivity κ ($\text{W m}^{-1} \text{K}^{-1}$)	1.5	Body side heat transfer h_h ($\text{W m}^{-2} \text{K}^{-1}$)	50	Air temperature T_c ($^{\circ}\text{C}$)	27

$\text{FF} = 0.25$, $L^* = \frac{L}{\text{FF}}$), while the optimum engineering leg length L_{opt}^* falls into a range of 3–6 cm under typical heat transfer coefficients, which is an order of magnitude larger than the real value. In fact, even though the optimum operating conditions cannot be realized, there still exists much room for improvement for the power density. With the leg length increasing from 1 mm to 2 mm, and the fill factor decreasing from 0.25 to 0.2, L^* could increase from 4 mm to 10 mm, resulting in a doubled power density (from $4.7 \mu\text{W cm}^{-2}$ to $9.7 \mu\text{W cm}^{-2}$) with an unchanged heat transfer coefficient, as illustrated by the dashed arrow in Fig. 5(a). When further increasing the hot and cold side heat transfer coefficients by 50%, which could be achieved by using better thermal interfacial materials or spreaders, the power density could realize a level of $19 \mu\text{W cm}^{-2}$, as shown by the dotted arrow, which means a rise by 3 times compared to the initial case.

To validate the above predictions, two TEGs were made with different geometric parameters for experimental verification (Fig. 5(b)), and their power densities (based on the substrate area) were measured under real heat transfer arrangements as

shown in Table 4. During all measurements, the tested TEG was placed on a heating panel in a wind channel, as shown in Fig. 5(c). The panel and ambient temperatures were kept constant, and the air velocity was fixed at 2 m s^{-1} , leading to a cold side heat transfer coefficient of about $50 \text{ W m}^{-2} \text{K}^{-1}$. In the former two tests (L^*h -1 and L^*h -2), a piece of thermal insulating foam with a thickness of 0.5 mm was fixed between the heating panel and the TEG, while in the third test (L^*h -3) the foam was removed, with the TEG being connected to the panel by a layer of double faced adhesive tape ($\sim 80 \mu\text{m}$ in thickness). The above two arrangements at the hot side respectively represent the normal and enhanced heat transfer conditions, and heat transfer coefficients of 50 and $150 \text{ W m}^{-2} \text{K}^{-1}$ are approximately given. Fig. 5(d) presents the measured power densities in the three tests as well as the corresponding model predictions based on eqn (16). With the engineering leg length L^* increasing from 5.8 to 10.3 under the same thermal circumstance, the power density increased from 4.2 to $6.5 \mu\text{W cm}^{-2}$. In the third test, owing to the enhancement of the hot side heat transfer, the power density realized a further increase to $14.1 \mu\text{W cm}^{-2}$, which means a rise by 2.4 times compared to the

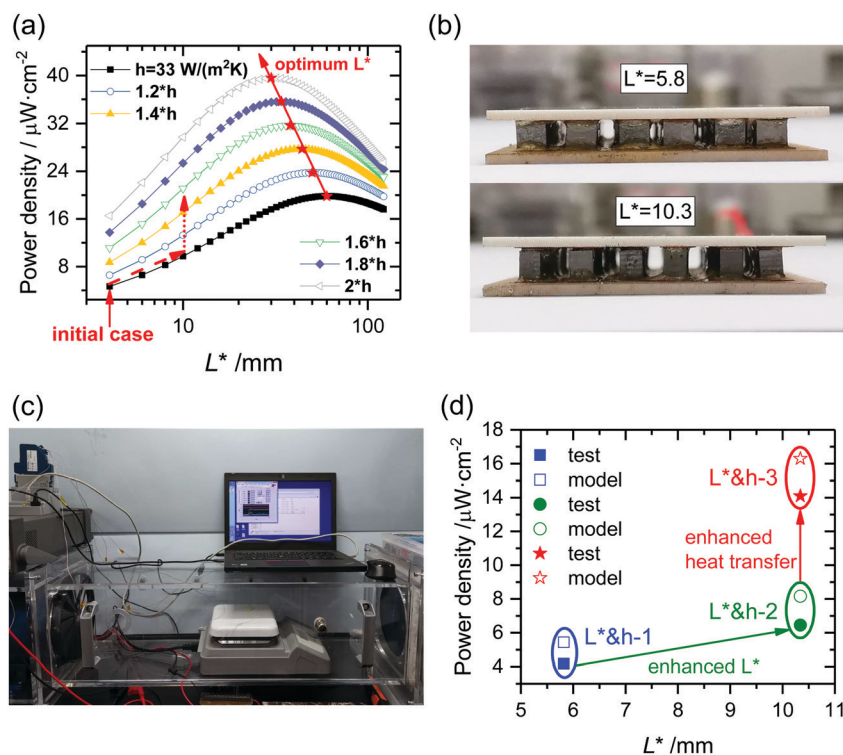


Fig. 5 (a) Variations of the power density with the engineering leg length and heat transfer coefficient, by analytical calculation; (b) pictures of two TEGs with different geometric parameters; (c) a picture of the testing wind channel; and (d) evolutions of the power density with enhanced engineering leg length and heat transfer coefficient, by both experiments and analyses.

Table 4 Geometric parameters and thermal conditions of the tested TEGs

Test no.	Leg length (mm)	Leg area (mm ²)	Fill factor	L^* (mm)	Heat source (°C)	Heat sink (°C)	Heat transfer coefficients (W m ⁻² K ⁻¹)
L*&h-1	1.78	4	0.30	5.8	32	23.7	50 (hot)/50 (cold side)
L*&h-2	1.9	2.4	0.18	10.3	32	23.7	50/50
L*&h-3	1.9	2.4	0.18	10.3	32	23.7	150/50

value in the first test. Reasonable agreements are achieved between the measured data and model predictions.

4.2 A general design procedure by power and voltage match

In addition to the power density, the output voltage density is another important index which should also be taken into account for a complete design process. A formula of the output voltage density under maximum power output conditions for given leg dimension parameters and heat transfer conditions is derived as

$$V_{L,\text{dens}} = \frac{1}{2} \alpha \Delta T \frac{rh}{L^*h + \kappa} \quad (22)$$

where $r = \frac{L}{A}$ is the aspect ratio of the TE leg. With eqn (16) and (22) at hand, here we present a general object-oriented design procedure for TEGs as shown in Fig. 6 and 7.

An object-oriented design of a TEG starts from the specifications of the target users' needs of output power and voltage (specified by its application), and the spatial layout and thermal conditions, from which the available temperature difference, area and the target power and voltage density of the TEG could be determined. Suitable thermoelectric materials could also be selected according to the temperature range. Once the target power density is settled, the engineering leg length L^* is accessible under designed heat transfer conditions according to eqn (16), as shown in Fig. 7(a). The fill factor and the leg length are given by a comprehensive trade-off among the device stability, contact effects, cost and weight *etc.* (Fig. 7(b)). With known target voltage density ($V_{L,\text{dens}}$) and L^* , the aspect ratio is obtained by eqn (22) (Fig. 7(c)). Finally, with the given leg length (L) in Fig. 7(b) and the aspect ratio in Fig. 7(c), the leg area is determined in Fig. 7(d). The target output power and voltage are expected to be achieved with the above designed

parameters and the optimum electrical operating condition given by eqn (15(d)).

It is worth noting that the above analysis ignores the secondary thermal effects within a TEG system, including the thermal spreading resistance within the substrate, the thermal shunt effect of air surrounding the TE legs and the thermal radiation between the substrates at the hot and cold sides, while it is found to be valid for fill factors higher than 5%, which is a much lower value compared to typical fill factors in current devices, implying a solid reliability of our model and design procedure for practical applications (see the ESI,† Design of TEGs with limited T , for detailed information).

5. Heat transfer effectiveness versus cost

At last, we perform several *in situ* thermal resistance estimations based on experimental or numerical data reported in the literature^{13–16,25,26,41–46} and present them in Fig. 8(a). The selected cases cover different combinations of heat transfer media at the hot and cold side as listed in the legend. In most cases the thermal resistance distributions in the TEG systems were not explicitly discussed, and we extract them from available temperature data including the total temperature ranges and the effective temperature ratios, as shown in Fig. 8(b). Most of the data points in Fig. 8(a) fall into the scope of 0.05–10, as is estimated in Section 3.1. Moreover, most TEG systems work under a total temperature difference of tens or hundreds of degrees, and the effective temperature ratio covers a wide range from 0.05 to 0.7. In other words, remarkable temperature loss universally occurs in practical TEG system applications, emphasizing the importance of thermal-electrical coupled optimizations of TEG systems.

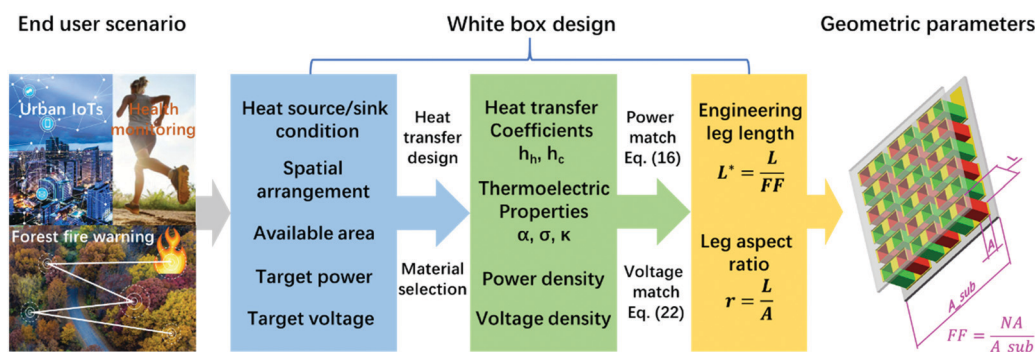


Fig. 6 A white box design of the TEG system by considering the power match and voltage match conditions.

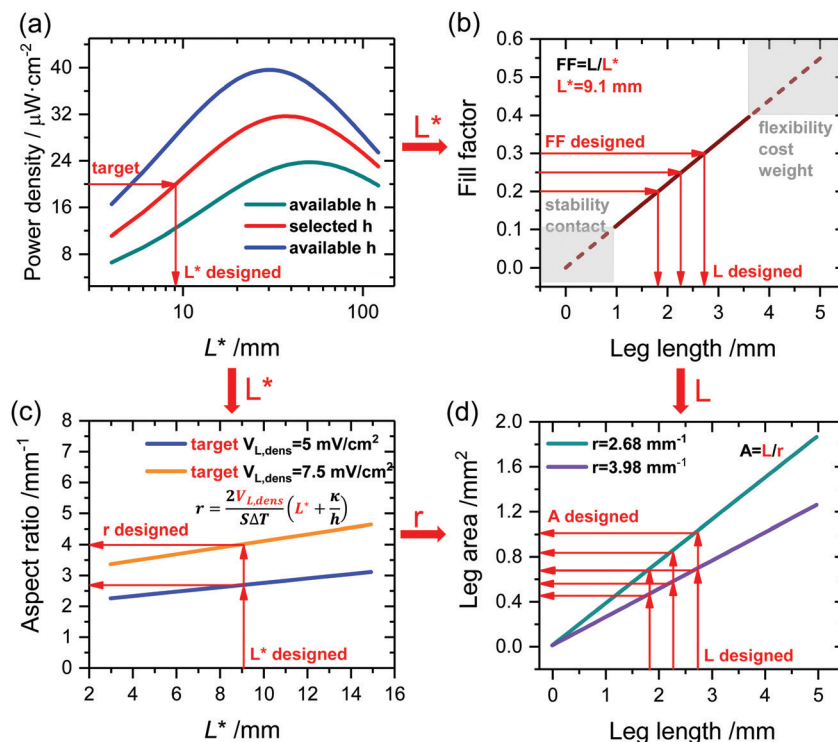


Fig. 7 An object-oriented design of a TEG. (a) Determination of L^* according to the target power density and heat transfer condition; (b) evaluation of the leg length and fill factor based on L^* ; (c) location of the aspect ratio according to the target voltage density and L^* ; and (d) determination of the leg area by the aspect ratio and the leg length.

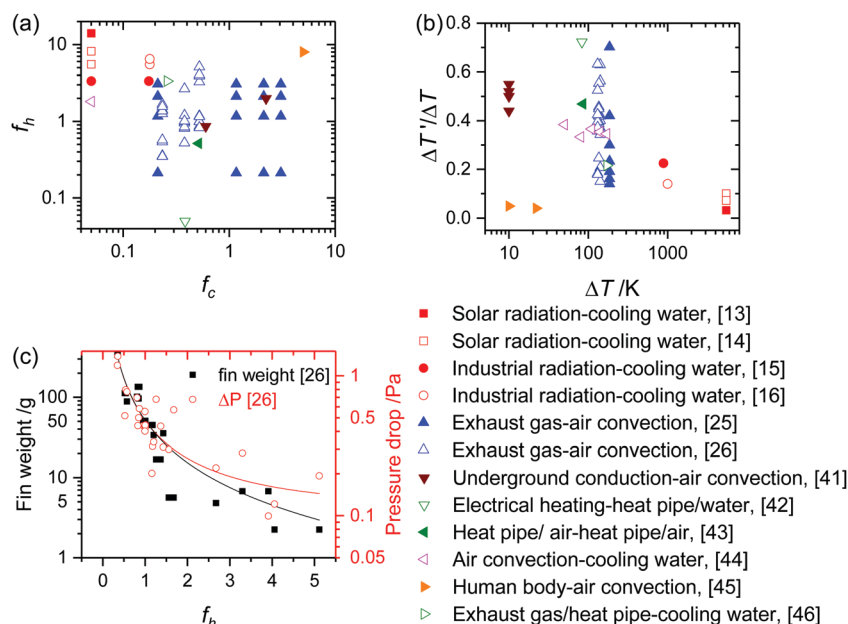


Fig. 8 (a) Thermal resistance ratio distributions and (b) temperature ranges and effective temperature ratios of various types of TEG systems; and (c) thermal management cost of a TEG system versus heat transfer performance.

In Fig. 8(c) we present variations of the fin weight and pressure drop per TE module for a chimney TEG system versus the hot side thermal resistance ratio, with the raw data extracted from ref. 26, in which a TEG system is arranged on

the side wall of the chimney, and a plate-fin heat exchanger is mounted on the hot side of the TEG system to extract heat from the exhaust gas. To enhance the heat exchange between the exhaust gas and the heat exchanger, a fan is used to improve

the gas velocity through the heat exchanger. The fin weight of the heat exchanger and the pressure drop through the fins can be respectively regarded as a direct reflection of the initial and operating costs of the thermal management of the TEG system. It shows that with an improvement in the heat transfer condition (a decrease of the hot side thermal resistance ratio), both the initial cost and the operating cost experience a sharp rise. This trend is also expected to work on other kinds of thermal management methods. This is the primary reason why diverse external thermal resistance levels exist in practical applications. In other words, the heat transfer coefficients between a TEG and the heat source (sink) could not be selected to be as high as possible. Instead, a system design should be done under a rational balance between the heat transfer effectiveness and cost, and also other potential factors such as space and weight.

6. Conclusions

In practical applications, thermoelectric power generators need to account for the effects of external thermal resistances, which could significantly reduce the effective temperature difference and the output performance. We have developed an analytical model for the system level analysis of a thermoelectric power generator, and proposed a set of explicit formulas for quickly evaluating the optimal operating conditions and the corresponding system efficiency and power output. In comparison to the ideal thermoelectric power and efficiency formulae, the new formulae account for both the material intrinsic properties and external thermal conditions, and are validated to give accurate predictions over wide temperature ranges and thermal resistance distributions. It is found that the thermal resistance on the cold side has a greater impact on the system performance compared to that on the hot side. The update of the electrical match condition in realistic thermal circumstances results in a 10–20% rise in the system power output in comparison to the conventional way. A general object-oriented design procedure for TEGs is also discussed based on the target users' needs of output power and voltage, as well as the spatial layout and external thermal conditions. This work provides a deep perspective on TEG system level design, bridging the device and system of thermoelectric power generation.

Conflicts of interest

There are no conflicts of interest to declare.

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